SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

ICT162
Object Oriented Programming

Tutor-Marked Assignment

July 2025 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 24 % of the final mark for ICT 162 Object Oriented Programming.

The cut-off date for this assignment is **Sunday, 19 October 2025, 2355 hours**.

Note to Students:

You are to include the following particulars in your submission: Course Code, Title of the TMA, SUSS PI No., Your Name, and Submission Date.

All TMAs must be submitted electronically through Canvas. No partial submission of TMA will be accepted unless otherwise specified.

Assignment Requirements:

- Unless specified in the question, you CANNOT use packages not covered in this module, e.g., re, collections, numpy, pandas etc.
- All classes and functions must be documented. Provide sufficient comments to your code and ensure that your program adheres to good programming practices such as not using global variables.
- Do not use the exit() function.
- Failing to do so can incur a penalty of as much as 50% of the mark allotted.

Submission Details:

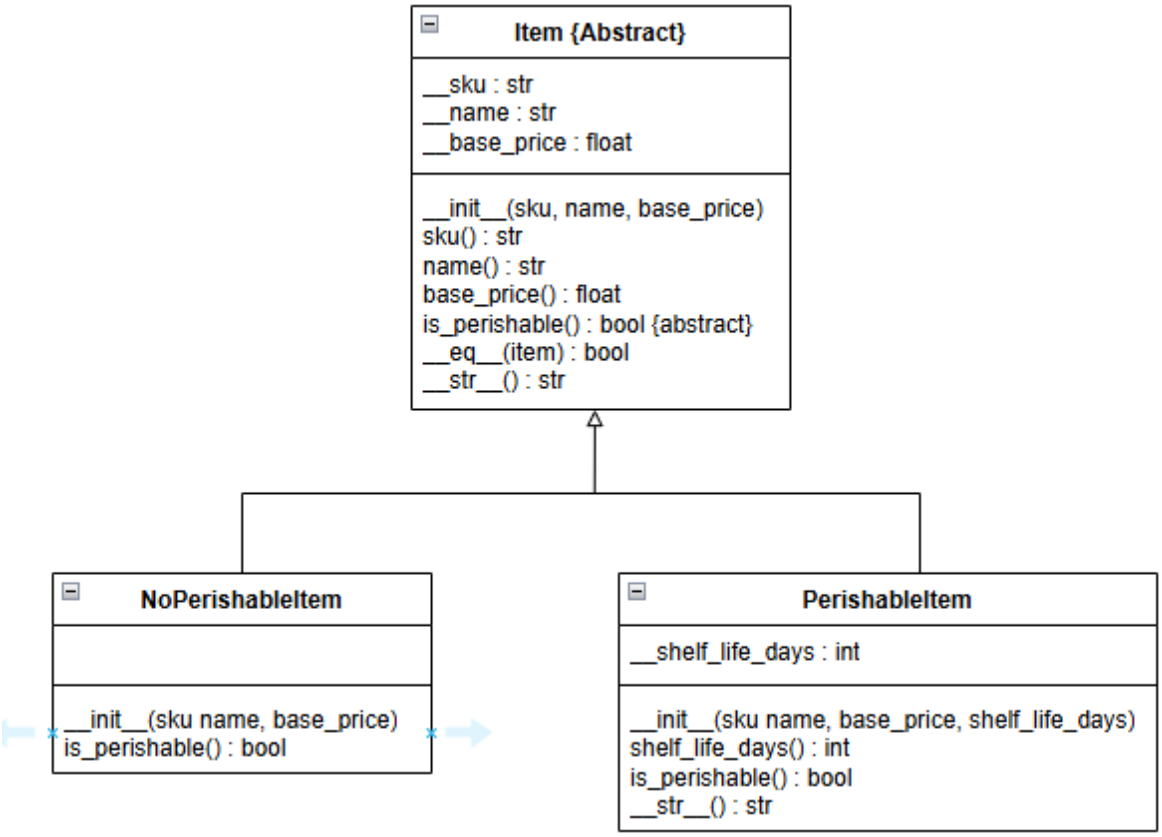
- Use the template word document provided - SUSS_PI_No-FullName_TMA.docx. Rename the file with your SUSS PI and full name join with "_TMA" e.g., "SUSS_PI-TomTanKinMeng_TMA.docx" (without the quotes).
- Include the following particulars on the first page of the word document, on separate lines: Course Code, SUSS PI No., Your Name, Tutorial Group and Submission Date.
- Copy and paste the source code of each program you write in text format. Submit screenshots for only output of your program, where applicable.
- If you submit the source code as a screenshot, your code will not be marked. That is, screenshot code will be awarded zero mark.
- Submit your solution in the form of a single MS Word document. Do NOT submit as a pdf document. You will be penalised if you fail to submit a word document.
- The word document must be submitted to your respective T group. Besides this, you are required to upload your source code to Vocareum.

Project Overview

You will build, step by step, a small warehouse and order fulfillment system using object-oriented design. The four questions build on each other and cover inheritance, composition, abstract classes, and exception handling. By the end, you will have a working text-menu program that loads synthetic data and processes orders with FEFO (first-expire, first-out) allocation and pricing rules.

Question 1 (20 marks)

The following class diagram shows 3 classes – Item, PerishableItem and NonPerishableItem class.



Implement an Abstract Item class and two concrete subclasses with validation.

The Item class is an abstract class with sku (str) using format XXX999, 3 uppercase letters followed by 3 digits, name (string) and base_price (float) as instance variables

- __init__: sku must be in the correct format (XXX999); name is non-empty after strip; base_price >= 0. On invalid, raise ValueError with a clear message.
- Define property for the instance variables.
- Define abstract is_perishable method.
- __eq__(item) compares with a similar item and returns True if the sku is the same, otherwise returns False. Raise Validation error if the parameter is not of Item type.
- __str__() returns a string in the following format:

```
SKU: A12B3C, Name: Milk, Base Price: 2.55
```

NonPerishableItem class is a subclass of Item. No additional instance variables.

- is_perishable(self) returns False

PerishableItem class is a subclass of Item. It has an additional shelf_life_days as instance variable. Shelf_life_days must be > 0, if not raise ValueError.

- __init__ then validate shelf_life_days > 0 (int). Raise ValueError if not.
- is_perishable() method returns True
- The __str__ method returns a string in the following format:

```
SKU: A12B3C, Name: Milk, Base Price: 2.55, Shelf Life: 7 days
```

Test the class by creating a PerishableItem and a NonPerishableItem and print the detail information of each product.

Example

```
# For a perishable item:
milk = PerishableItem('ABC333', 'Milk', 2.55, 7)
```

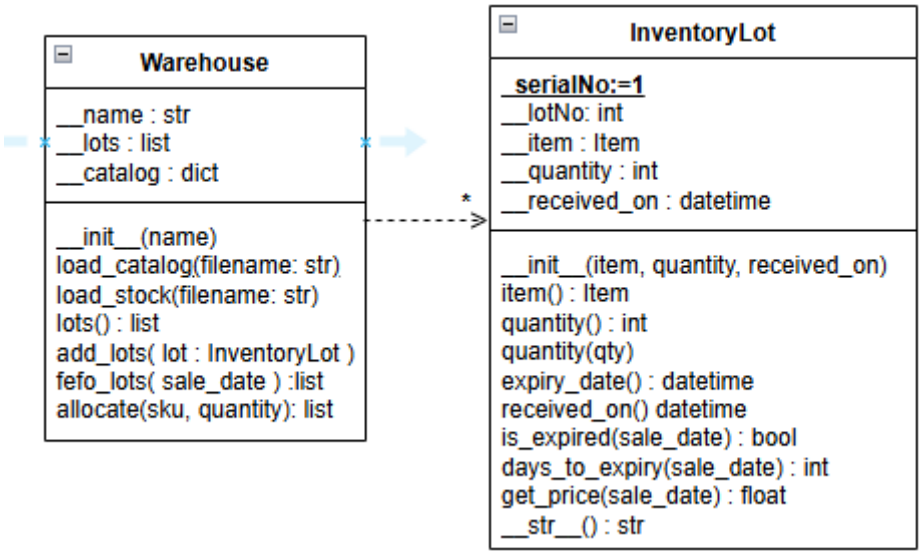
Output:

```
SKU: ABC333, Name: Milk, Base Price: $2.55, Shelf Life: 7 days
```

(20 marks)

Question 2 (25 marks)

Build inventory structures composing of lots in a Warehouse. The following is the class diagram.



The InventoryLot class links an Item to a quantity and received_on date.

The constructor has 3 parameters – item, quantity and received_on. Raise appropriate exceptions, e.g. must be an Item, positive quantity and appropriate date(cannot be earlier than

today). The __lotNo is assigned the running consecutive numbers from _serialNo class variable.

Implement property for the 3 instance variables and setter for the quantity.

The expiry_date() method returns the expiry date (received_on date + shelf life) of the item if the item is perishable, otherwise return None.

The is_expired(sale_date) returns True if the item is perishable and sale_date is later than expiry date, otherwise return False

The get_price(sales_date) method checks if the item is perishable/non-perishable. If it is not perishable, it returns the base price. For perishable items, pricing is lot-sensitive based on days to expiry at sale time:

- If sale_date > expiry_date: cannot sell this lot (expired). Raise ExpiredItemError() with an appropriate message.
- If 0 < days_to_expiry <= 2: 50% of base_price.
- If 2 < days_to_expiry <= 5: 80% of base_price.
- Else: 100% of base_price.

The __str__() method returns a string containing information about the inventory lot as follows:

```
Lot 12 SKU: BHC233 Name: Milk Base price: $4.50 Qty: 90 Received: 2025-05-31, Expires 2026-10-5
```

Expiry shown only for perishable items.

(25 marks)

Question 3 (25 marks)

The Warehouse class has 3 instance variables: the name of the warehouse, a dictionary of catalog of items and a list of InventoryLot.

Implement property for lots and catalog. The add_lots(lot) method adds InventoryLot lot to the lots list.

Method load_catalog(filename) reads data from given filename and builds a dictionary of a catalog of items. The key is sku and value are the Perishable or NonPerishable objects. Assume data contents are correct.

Method load_stock(filename) that reads data from given filename and builds a list of InventoryLot objects. Assume data contents are correct.

The fefo_lots(sale_date) method will sort all the lots according to first by expiry date, then received_on, then sku. All nonperishables are listed behind the perishables. Return this list of

items. Implement FEFO ordering (earliest expiry first; non-perishables always after perishables). Ties break by received_on (older first), then by sku label.
The allocate(sku, quantity) method implements the following:

- Attempt to allocate the sku with a certain quantity to fulfill an order. E.g. 65 units of ABC100 are ordered. The 'allocate' method will allocate 60 units from lot x received on June 6, expiring on June 8 and 5 unit from lot y received on June 7, expiring on June 9, assuming today is June 7.
- The 'allocate' method will first try to decide if the order can be fulfilled, if not OutOfStockError will be raised.
- Raise InvalidSKUError if sku is not known to exist anywhere in the warehouse.
- Skip expired lots; if all lots are expired and the item is perishable, raise OutOfStockError with message showing available lot is 0.
- If at least one lot would be expired on sale_date and attempted to be priced/allocated, raise ExpiredItemError that indicates the first offending lot's details.
- Quantity of the lot should be updated ONLY if order is fulfilled, and if it becomes zero, the inventory lot should be removed from the list.

Example

```
bread = PerishableItem("BRD123", "Bread", 1.40, 2)
beans = PerishableItem("BN123", "Beans", 1.00, 10)
milk = PerishableItem("MIL123", "Milk", 2.10, 3)

lot1 = InventoryLot(bread, 5, datetime(2024,6,2))
lot2 = InventoryLot(beans, 50, datetime(2024,6,15))
lot3 = InventoryLot(milk, 10, datetime(2024,6,1))
```

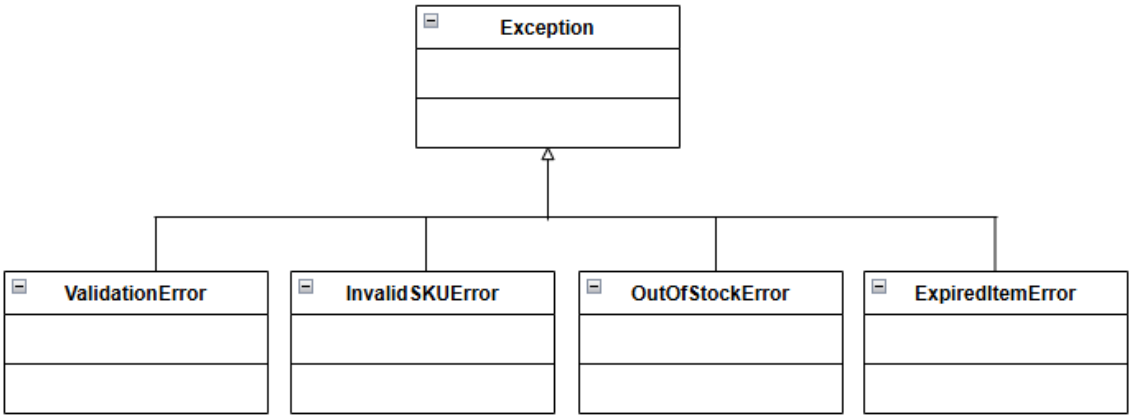
FEFO order:
Milk → Bread → Bean

(25 marks)

Question 4 (5 marks)

Introduce domain-specific exceptions, validate inputs thoroughly, and implement order allocation.

Design the following user defined exceptions and raise these specific exceptions in your implementation.



All constructor and mutating methods validate inputs and raise ValidationError (not ValueError) for domain validation (negative qty, empty id, invalid sku).

(5 marks)

Question 5 (25 marks)

Write a simple text-menu application that:

- Loads a catalog from cat.csv and initial stock from stocks.csv.
- Let the user list catalog and current inventory, restock, and place orders.
- Demonstrates exception handling by gracefully reporting errors.

Application menu requirements (i.e., what the main() function should be doing)

On start:

- Create Warehouse.
- Build catalog.
- Build stock.

Present loop:

1. List catalog

2. Show inventory
3. Restock
4. Place order
5. Quit

Option details:

1. List catalog: print sku, name, base_price, shelf life (for Perishable items).
2. Show inventory
    - Show all inventory lots.
3. Restock:
    - Prompt: sku, qty, received_on (YYYY-MM-DD).
    - Create and add InventoryLot; handle ValidationError and report issues.
4. Place order:
    - Prompt: sku; then qty.
    - On success: print receipt:
        - For each line: show components with lot received_on, expiry date and quantity fulfilled; then order total.
    - On failure: show OutOfStockError or InvalidSKUError message; no inventory change.
5. Quit: exit loop.

Error handling:

- All invalid inputs handled; do not crash. Show the exception type and message succinctly.

(25 marks)

Appendix:

List catalog

```
MENU:
1. List catalog
2. List inventory
3. Restock item
4. Place order
5. Exit
---- CATALOG ----
SKU: A12B3C Name: Milk Price: $2.55
SKU: ABC100 Name: Bread Loaf Price: $2.29 Shelf Life: 5 days
SKU: JKL309 Name: Milk Price: $1.49 Shelf Life: 7 days
SKU: PQR310 Name: Chicken Breast Price: $6.25 Shelf Life: 6 days
SKU: ZED888 Name: Apple Price: $0.80 Shelf Life: 12 days
SKU: BHC233 Name: Toilet Paper Price: $2.99
SKU: MZT400 Name: Butter Price: $2.50 Shelf Life: 10 days
SKU: JKM681 Name: Eggs (Dozen) Price: $2.99 Shelf Life: 14 days
```

List inventory

```
MENU:
1. List catalog
2. List inventory
3. Restock item
4. Place order
5. Exit
---- INVENTORY ----
Lot: 001 SKU: ABC100 Name: Bread Loaf Price: $2.29 Shelf Life: 5 days, Qty:
72 Received: 2025-06-03, Expires: 2025-06-08
Lot: 002 SKU: ABC100 Name: Bread Loaf Price: $2.29 Shelf Life: 5 days, Qty:
60 Received: 2025-05-29, Expires: 2025-06-03
Lot: 003 SKU: JKL309 Name: Milk Price: $1.49 Shelf Life: 7 days, Qty: 43
Received: 2025-06-05, Expires: 2025-06-12
Lot: 004 SKU: JKL309 Name: Milk Price: $1.49 Shelf Life: 7 days, Qty: 61
Received: 2025-06-02, Expires: 2025-06-09
Lot: 005 SKU: JKL309 Name: Milk Price: $1.49 Shelf Life: 7 days, Qty: 32
Received: 2025-05-28, Expires: 2025-06-04
Lot: 006 SKU: PQR310 Name: Chicken Breast Price: $6.25 Shelf Life: 6 days,
Qty: 27 Received: 2025-06-04, Expires: 2025-06-10
Lot: 007 SKU: PQR310 Name: Chicken Breast Price: $6.25 Shelf Life: 6 days,
Qty: 38 Received: 2025-06-01, Expires: 2025-06-07
Lot: 008 SKU: PQR310 Name: Chicken Breast Price: $6.25 Shelf Life: 6 days,
Qty: 41 Received: 2025-05-31, Expires: 2025-06-06
Lot: 009 SKU: ZED888 Name: Apple Price: $0.80 Shelf Life: 12 days, Qty: 100
Received: 2025-06-02, Expires: 2025-06-14
Lot: 010 SKU: ZED888 Name: Apple Price: $0.80 Shelf Life: 12 days, Qty: 78
Received: 2025-06-04, Expires: 2025-06-16
Lot: 011 SKU: ZED888 Name: Apple Price: $0.80 Shelf Life: 12 days, Qty: 65
Received: 2025-05-30, Expires: 2025-06-11
Lot: 012 SKU: BHC233 Name: Toilet Paper Price: $2.99, Qty: 90 Received:
2025-05-31, Expires: --
```

```
Lot: 013 SKU: BHC233 Name: Toilet Paper Price: $2.99, Qty: 87 Received:
2025-06-03, Expires: --
Lot: 014 SKU: BHC233 Name: Toilet Paper Price: $2.99, Qty: 92 Received:
2025-06-05, Expires: --
Lot: 015 SKU: MZT400 Name: Butter Price: $2.50 Shelf Life: 10 days, Qty: 44
Received: 2025-05-31, Expires: 2025-06-10
Lot: 016 SKU: MZT400 Name: Butter Price: $2.50 Shelf Life: 10 days, Qty: 62
Received: 2025-06-01, Expires: 2025-06-11
Lot: 017 SKU: MZT400 Name: Butter Price: $2.50 Shelf Life: 10 days, Qty:
51) Received: 2025-06-04, Expires: 2025-06-14
Lot: 018 SKU: JKM681 Name: Eggs (Dozen) Price: $2.99 Shelf Life: 14 days,
Qty: 39 Received: 2025-06-02, Expires: 2025-06-16
Lot: 019 SKU: JKM681 Name: Eggs (Dozen) Price: $2.99 Shelf Life: 14 days,
Qty: 49 Received: 2025-05-28, Expires: 2025-06-11
Lot: 020 SKU: JKM681 Name: Eggs (Dozen) Price: $2.99 Shelf Life: 14 days,
Qty: 58 Received: 2025-06-06, Expires: 2025-06-20
```

Place order (65 qty of ABC100, assuming the sale date is 06-01)

```
MENU:
1. List catalog
2. List inventory
3. Restock item
4. Place order
5. Exit
Allocated 60 from lot 002 received 2025-05-29, expires 2025-06-03
Allocated 5 from lot 001 received 2025-06-01, expires 2025-06-06
Order fulfilled: 65 units of ABC100 (Bread Loaf)
```

---- END OF ASSIGNMENT ----